

Can't find the truth in AI search?

Nuclear engineers spend **4–8 hours** manually searching documents
for a single compliance question.

AI hallucination in regulated environments isn't a bug — it's a
liability.



PulseDNA

THE PROBLEM

Nuclear Compliance Is Broken.

The industry's most critical decisions are still made by engineers drowning in documents — without AI they can trust.

4–8 wks

Typical safety case preparation per cycle

100–300 hrs

Manual traceability mapping per submission

2–5 days

Change impact analysis per request

15 sec

Magic Lens ECCS compliance package

\$5–15M

PulseFINE™ full infrastructure build



AI Hallucination Risk

Generic LLMs cite non-existent regulations. In nuclear, a wrong citation isn't a typo — it's a reportable event.



Document Search Overload

Engineers manually cross-reference NUREG documents, NRC ADAMS, and national lab readouts for every submission.



No Audit Trail

Current tools can't prove how a compliance conclusion was reached — creating regulatory liability at every review.



Data Sovereignty Gaps

Public AI tools expose sensitive operational data to cloud infrastructure — unacceptable for nuclear-grade environments.



Licensing Delays Cost Billions

NRC licensing bottlenecks delay SMR deployment by years. Every week of delay is a week of lost clean energy capacity.



No Predictive Intelligence

Facilities react to anomalies instead of predicting them. Without edge AI, early-warning signals are invisible until it's too late.

Two Products. **One** Compliance Journey.

Start with compliance acceleration today. Scale to operational sovereignty when you're ready.



PHASE 1 · ENTRY POINT

Magic Lens

A federated AI research co-pilot built for nuclear compliance engineers. Surfaces regulatory citations, cross-references NUREG documents, and assembles LER submission packages in seconds — directly in your browser.

- ✓ 10 CFR Part 50 citation engine with NUREG-0800 cross-reference
- ✓ Automated ECCS compliance package assembly in under 15 seconds
- ✓ Full provenance chain on every output — document ID, version, timestamp
- ✓ Chrome extension — no infrastructure, no IT procurement, live in days
- ✓ Federated Research Network routing to ORNL, INL, Sandia, PNNL

\$50K – \$200K / yr

Days to weeks



PHASE 2 · FULL INFRASTRUCTURE

PulseFINE™

Federated Intelligence for Nuclear Engineering. Purpose-built for SMRs — real-time anomaly detection, dual simulation workflows, and cross-site forecasting without centralizing sensitive data or relying on public cloud AI.

- ✓ Edge-resident AI agents monitoring reactor telemetry in real time
- ✓ Federated forecasting across SMR sites without sharing raw operational data
- ✓ GPU-optimized inference (Supermicro bare-metal + NVIDIA) for operational workloads
- ✓ PulseTRACE: model lineage, decision rationale, immutable audit logs
- ✓ Full air-gap capability — zero cloud dependency — zero data exfiltration risk

\$5M – \$15M Build

Months to multi-year



Key Partners: Supermicro (sovereign hardware) · Hadron Energy (prime reactor developer) · John Snow Labs (unstructured clinical & regulatory data) · NVIDIA · Google AI Futures Fund

THE TEAM

The Minds Behind Magic Lens

Problems come looking for you — when you have the experience to solve them.



Stephanie Joyce

CEO & FOUNDER

Ex-Apple · DOE R&D · MIT Research. Saw the pattern: nuclear compliance is the last frontier untouched by verified AI. Built PulseDNA to close that gap.

Apple

DOE R&D

MIT Research



Eugene Jin

CTO

Google · Meta · Oracle. Nuclear Engineering + AI. Architected the federated intelligence layer that makes verified, sovereign AI possible at reactor scale.

Google

Meta

Nuclear Engineering



Loic Fourniller

COO

3× Deep Tech COO · \$100M+ exit. Scaled systems from vision to revenue. Brings the operational rigor to turn breakthrough AI into enterprise-grade deployments.

3× COO

\$100M+ Exit

Deep Tech

"We are not building a search engine. We are building a co-processor for human reasoning — one that ensures every decision made at a nuclear facility is grounded in the best available verified knowledge. Magic Lens gets you there in weeks. PulseFINE™ gets you there forever."

— Stephanie Joyce, Founder & CEO, PulseDNA

Verified Intelligence at the Point of Need

From NRC compliance acceleration to sovereign operational AI — PulseDNA meets nuclear operators where they are today.



PHASE 1 · MAGIC LENS

NRC Compliance Acceleration

- ✓ ECCS compliance package assembled in <15 seconds (vs. 4–8 hours manually)
- ✓ 10 CFR Part 50, NUREG-0800, NUREG-1022 citation engine built-in
- ✓ LER and CAR package automation with full provenance chain
- ✓ NRC ADAMS cross-reference and precedent LER surfacing
- ✓ Credentialed SME routing via Federated Research Network
- ✓ Chrome extension — no IT procurement, live in days

\$50K–\$200K / yr

⚡ Days to weeks



PHASE 2 · PULSEFINE™

Sovereign Operational AI

- ✓ Real-time anomaly detection on reactor & infrastructure telemetry
- ✓ Early-warning signals before out-of-spec events occur
- ✓ Dual simulation: predictive maintenance + operational scenario modeling
- ✓ Federated forecasting across SMR sites — no raw data sharing
- ✓ PulseTRACE immutable audit logs for NRC and DOE oversight
- ✓ Full air-gap capability — Supermicro + NVIDIA bare-metal

\$5M–\$15M Build

🏗️ Months to multi-year



NRC Compliance Officers

LER & CAR package assembly in seconds

Magic Lens



SMR Operators

Real-time anomaly detection across sites

PulseFINE™



Chief Nuclear Officers

AI-assisted compliance documentation

Magic Lens



National Labs & DOE

Sovereign AI — ORNL, INL, Sandia, PNNL

PulseFINE™

Two Products. One Compliance Journey. Built for Nuclear-Grade Operations.

Start with **Magic Lens** — the AI research co-pilot that accelerates NRC compliance documentation in weeks.

Scale to **PulseFINE™** — the full federated AI infrastructure for anomaly detection, dual simulation, and sovereign intelligence at the edge.

 **Magic Lens** — Entry Point

 **PulseFINE™** — Full Infrastructure

 **Air-Gapped · Zero Public Web**

 **10 CFR Part 50 · NUREG-0800**

 [pulsednafederatedai.com/ nuclear](https://pulsednafederatedai.com/nuclear)

Explore Magic Lens →

Explore PulseFINE™ →

AFFILIATIONS & RECOGNITIONS

SAM.GOV

SBA

IEEE NPSS

U.S. Dept. of Energy

Pacific AI

World Inst. for Nuclear Security